MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 12BE6

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Pentagrid converter, miniature

Outline	---	5-2 (EIA)
Base	---	E7-1
Envelope	---	T5-1/2
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7
Element	---	g1	k, g5	h	h	a	g2 and 4	g3

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec2 and 4	Ec1	Ec3	Pp	Pg2 and 4	Ic1	Ik	(B) Ehk	(B) Alt
Unit:	V	Vdc	Vdc	Vdc	Vdc	W	W	mAdc	mAdc	v	ft
Maximum:	13.9	330	110	---	---	1.1	1.1	0.5	15.5	100	(See note 1)
Minimum:	11.3	---	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	12.6	250	100	0	-1.5	---	---	---	---	---	---

GENERAL:

Qualification - Not required

(B) denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1326	Conversion transconductance (1)	See note 2	0.65	II	Sc	280	660	µmhos
1266	Total grid current	Ec3 = -2 Vdc (see notes 2 and 3)	0.65	II	Ic3	0	-2.0	µAdc
1256	Electrode current (1) (anode)	See note 2	0.65	II	Ib	1.9	4.1	mAdc
1256	Electrode current (2) (anode)	Ec1 = -14 Vdc (see note 4)	0.65	II	Ib	0	50	µAdc
1256	Electrode current (screen)	See note 2	0.65	II	Ic2 and 4	5.2	9.8	mAdc
1231	Emission	Eb = Ec1 = Ec2 = Ec3 = 15 Vdc (see note 3)	0.65	II	Is	50	---	mAdc
(B) 1201	Short and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1031	Low-frequency vibration	Rp = 10,000 ohms; Ec1 = Ec3 = -1.5 Vdc	---	---	Ep	---	300	mVac
1301	Heater current		---	---	If	138	162	mA
(B) 1336	Heater-cathode leakage		---	---	Ihk	---	20	µAdc
1326	Conversion transconductance (2)	Ec3 = -30 Vdc (see note 2)	---	---	Sc	1	50	µmhos
1326	Conversion transconductance (3)	Ec3 = -6 Vdc (see note 2)	---	---	Sc	60	300	µmhos
1306	Oscillator transconductance	See note 4	---	---	Sm (g1-g2)	5,500	9,000	µmhos
1316	Oscillator amplification factor	See note 4	---	---	Mu	17	25	---
1256	Electrode current (cathode)	See note 4	---	---	Ik	16	33	mA
1331	Direct-interelectrode capacitance	Shield No. 316	---	---	Cg3-p Cg3-all Cp-all	--- 5.5 11.0	0.35 8.5 16.0	pF pF pF
(B) 1121	Base strain		---	---	---	---	---	---
(B) 2126	Glass strain		4.0	I	---	---	---	---
(B) 1211	Insulation of electrodes		4.0	S3	---	---	---	---
(B) 1105	Permanence of marking		---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						MIN	MAX	
	Quality conformance inspection, part 3							
---	Life-test provisions	Group A; Ihk = 100 V (see note 5)	---	---	---	---	---	---
---	Life-test end points (500 hours)	Conversion transconductance (1)	---	---	Sc	250	---	µmhos
		and Oscillator transconductance	---	---	Sm (g1-g2)	4,700	---	µmhos

NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage.
2. Insert 20,000 ohms from grid No. 1 to ground. Apply sufficient ac signal to produce 0.5mAdc average grid current.
3. This test to be performed at the conclusion of the holding period.
4. Nonoscillatory condition. Eb = 100 Vdc; Ec3 = 0; tie screen and anode together. Tie g3 to cathode.
5. Ec1 =-16 Vdc; Ec3 = 0; apply 16.5 Vac to grid No. 1. Ec1 may be obtained by self rectification.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - EL
Navy -
Air Force - 11, 80
DSA - ES

User activities:
Army - WC
Navy - AS, OS, CG, SH
Air Force - 19

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2425-32)

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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